Universidade de Fortaleza (UNIFOR)
Fortaleza, Ceará, Brazil
naubergois@gmail.com

August 10, 2026

Editorial Office
Ecological Informatics
Elsevier

Dear Editor,

We submit the manuscript entitled:

Ceará Wildfire Digital Twin: An Open-Source Agentic AI Platform with LangGraph

for consideration as a research article in *Ecological Informatics*.

Summary. Wildfires threaten biodiversity and ecosystem services in Brazil's semi-arid Caatinga, yet operational monitoring still relies on centralized satellite products with multi-hour latency. We present an open-source ecological data platform that assimilates NASA FIRMS and Open-Meteo within a ten-node LangGraph pipeline (ingestion, validation, parallel geospatial/climate analysis, evidence fusion, risk classification, ReAct diagnosis, and three-class alerts) for sustainable fire management.

The manuscript combines (i) computational ecology tools for multi-source ecological observation streams, (ii) a three-class alert framework (NO/UNCERTAIN/YES) with explicit precision–recall trade-offs validated against INPE, and (iii) NeKo-PIGNN (Neural Koopman Operator + Physics-Informed GNN) as an offline research module for physics-regularized prediction benchmarks.

Key Results.

- **Dry-season ecological alerts:** 84.2% precision and 91.8% recall over the 2025 dry season (184 days, 24,573 INPE hotspots); bootstrap 95% CI for YES precision.
- **Transferability:** under the same protocol in Maranhão and Piauí, recall > 96% at 58–60% precision, motivating per-state calibration.
- **NeKo-PIGNN:** $R^2 = 0.972$ on synthetic dynamics; real-data limits reported separately from operational alerts.
- **Reproducibility:** MIT-licensed code and experiment scripts on GitHub (https://github.com/naubergois/ceara-queimadas).

Fit with the Journal. *Ecological Informatics* publishes computational ecology and ecological data science for ecosystem analysis and sustainable management. Our contribution targets novel tools for monitoring and synthesizing fire-related ecological data in a biodiversity-sensitive dry forest biome, with machine-learning and agentic pipelines that support management decisions under climate-driven drought. We believe the Caatinga case study will interest readers working on ecological remote sensing, fire regimes, and open ecological software.

Prior submission. A related version was previously submitted to *Environmental Modelling & Software* (manuscript ENVSOFT-D-26-02266) and was not accepted for that venue. The present manuscript has been revised for Ecological Informatics scope (ecological framing, length, and journal metadata) and is not under consideration elsewhere.

Suggested Reviewers.

1. Prof. Steven L. Brunton — University of Washington (Koopman operators, data-driven dynamical systems)
2. Prof. George Em Karniadakis — Brown University (physics-informed machine learning)
3. Prof. Alan A. Ager — USDA Forest Service (wildfire spread modelling and operational tools)

Declarations. All five authors (Nauber Gois, Alexandre Lobo, Vladia Celia Monteiro Pinheiro, Daniel Valente de Macedo, Adriano Bessa Albuquerque) approved the manuscript. We declare no competing interests. The study uses publicly available satellite and meteorological data; no human subjects were involved. *Ecological Informatics* is a full open-access journal; we acknowledge the APC policy.

Sincerely,

Nauber Gois
On behalf of all co-authors